Mohamed Mahmoud

TECHNICAL LEAD / SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Lead with 7+ years building production systems across backend, web and mobile, with Laravel (PHP) as my core backend stack. At Tripklik I lead engineering on a travel booking platform serving 1M+ users a month and 5,000+ searches and orders, and take part in its architecture across Python (FastAPI) and Laravel services, Next.js web apps and React Native apps: concurrent multi-supplier search with background workers and polling, indexing and serialization work that made hotel and flight results about 4× faster and API responses about 5× faster, an order state machine, and a multi-tenant design where adding a new agency or brand is configuration rather than new code. I also built the end-to-end authentication and permission flow and the platform's AI integrations (MCP for Claude and ChatGPT, WhatsApp with Claude). I automate delivery end to end with Bitbucket Pipelines, Docker, Kubernetes, AWS and Vercel. I treat testing as part of the work: Laravel unit and feature tests with PHPUnit, Playwright end-to-end tests, and PHP and TypeScript linting with static analysis. Before that, led backend squads delivering telecom value-added services for operators in KSA, UAE and Kuwait.

CORE COMPETENCIES

Software Architecture & Design Patterns | Laravel & Backend APIs | Search Performance & Indexing | Concurrency, Queues & Polling | State Machines & Order Lifecycle | CI/CD & Deployment Automation | Authentication & Permissions | AI Integration (MCP)

WORK EXPERIENCE

Tripklik

Oct 2021 - Present

Technical Lead

Senior Software Engineer Oct 2021 - Oct 2024; Technical Lead since Oct 2024.
Riyadh, KSA (Remote)

- Lead engineering on a travel booking platform serving 1M+ users a month and 5,000+ searches and orders: architecture, code review and release decisions across backend, web and mobile.
- Search architecture: took part in designing concurrent search across all suppliers (Amadeus, Hotelbeds, Flyadeal) with background workers and queues; clients poll for results as each supplier responds, and circuit breakers isolate slow or failing suppliers so the search returns what arrived instead of failing.
- Search performance: hotel and flight results about 4× faster after adding indexing, polling and queue-based processing; result filtering in the web app 7× faster with indexed lookups and virtualized lists.
- API performance: took part in replacing SQLAlchemy-based JSON responses in the Python search service with a Rust-based serializer, making request/response about 5× faster with much lighter payloads.
- Hotel mapping: worked on the Central Station service that maps every hotel across all providers to a single record, so results from different suppliers merge and match quickly.
- Order lifecycle: moved order handling onto an explicit state machine (states and allowed transitions), and shipped B2B order operations end to end: void with fee preview, refunds, PNR retrieval and import, hold orders, invoices, balances and top-ups.
- Laravel booking and finance core: built ancillary products (eSIM, baggage protection, "blind choice" fares) behind feature flags and fixed flight and hotel markup calculations; worked on multi-currency conversion, agency wallets on ACID MySQL ledgers and a payment-gateway factory (Stripe, PayPal).
- Testing and code quality: Laravel feature and unit tests with PHPUnit (mocks and test data), end-to-end tests with Playwright, Laravel debugging and logs for production issues, and PHP and TypeScript linting with static analysis (PHPStan) to keep the code well structured.
- Deployment automation and CI/CD: Bitbucket Pipelines that build, test and deploy automatically; Docker and Docker Compose for services and local environments, Kubernetes for orchestration, and the deployment lifecycle on AWS servers; frontend apps deploy to Vercel automatically from the pipeline or straight from a branch.

- Multi-tenant scalability: one codebase serves the consumer site, the agency portal and several brands (tenant routing, per-brand configuration), so onboarding a new agency or brand needs configuration rather than new code, and teams can extend their areas independently.
- Authentication and permissions end to end: OTP sign-in, role-based permissions enforced in the API and at the edge, and scoped short-lived tokens for links coming from AI assistants.
- AI integration: built an MCP server connecting the booking engine to Claude and ChatGPT (live flight and hotel search with agency pricing, deep links back to book) and worked on a WhatsApp integration with Claude so agencies can continue a booking from chat.
- Led the modernization of the multi-brand mobile apps (Tripklik, Destinia): React Native 0.85 + Expo SDK 56 with EAS multi-brand builds, per-brand runtime configuration, and Kotlin and Swift build fixes.
- Set up AI-assisted engineering for the team: agent guides, reusable skills and safety rules derived from real incidents.

Ivas

Feb 2019 - Jun 2021

Senior Software Engineer

Cairo, Egypt

- Directed core backend development and infrastructure for Ringback Tone delivery platforms run by major telecom operators in KSA, UAE and Kuwait.
- Moved production backends from monolithic MVC to a clean layered architecture on Laravel and MongoDB to simplify operator integrations.
- Led backend squads on API security, performance profiling and maintainable delivery.

Grand

Jun 2018 - Dec 2018

Back-End Developer

Cairo, Egypt

- Built REST APIs for mobile and web apps with a focus on performance and security.
- Defined API contracts with frontend teams.

LoadServ For Integrated Business Solution

Jan 2018 - May 2018

Full-Stack Developer

Cairo, Egypt

- Delivered frontend and backend features on team projects; took part in code reviews.

EDUCATION

BSc Computer Science Helwan University

2014 - 2018

Cairo, Egypt

Graduation project: face-recognition platform for finding missing children (Python, Flask, OpenCV, Firebase).

SKILLS

Languages: TypeScript, JavaScript, PHP 8, Python, SQL

Backend: Laravel 8-10, FastAPI, Node.js, Express, Socket.IO, REST APIs, Webhooks, JWT, Sanctum, Swagger/OpenAPI

Frontend: React, Next.js, Redux Toolkit, RTK Query, Tailwind CSS, SCSS, Storybook, Vue 3, RTL / i18n

Mobile: React Native, Expo / EAS, Swift native modules, multi-brand builds **Data:** MySQL, Redis, MongoDB, Meilisearch, Supabase

Testing & Quality: PHPUnit, Laravel feature and unit tests, mocks and test data, Playwright (end-to-end), PHPStan, PHP and TypeScript linting

DevOps & Infrastructure: Docker, Docker Compose, Kubernetes, CI/CD (Bitbucket Pipelines), deployment automation, AWS (servers, S3), Vercel, Nginx, Cloudflare

Architecture: Microservices, event-driven architecture, multi-tenant / multi-brand SaaS, resilience patterns (circuit breaker, backoff), design patterns

AI: Model Context Protocol (MCP), Claude / ChatGPT integrations, AI coding agents, n8n